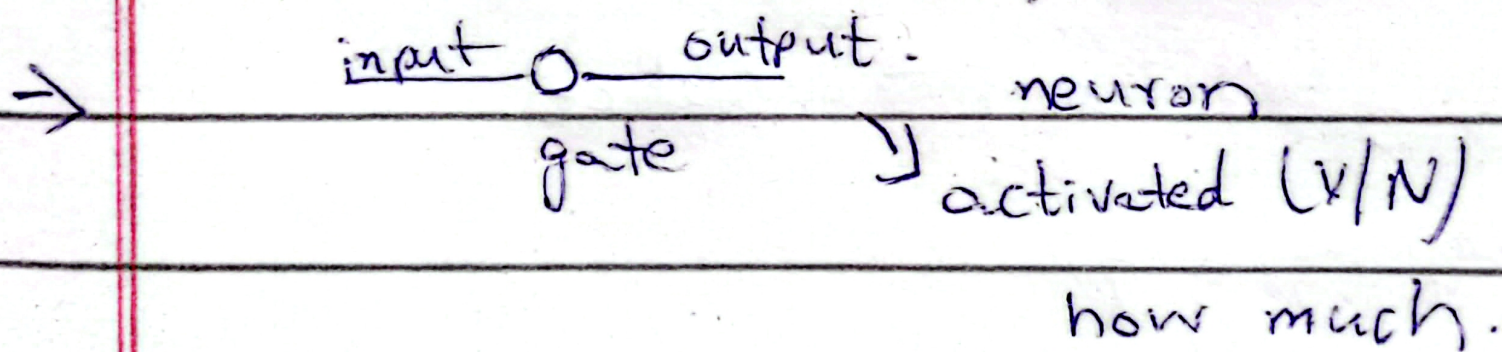


Activation func in neu. Net:-

In artificial neural network each neuron forms a weighted average of its inputs & passes resulting scalar value through a function referred to as act. function.



Sigmoid $\rightarrow [0, 1]$

$\tan 1 \rightarrow [-1, 1]$

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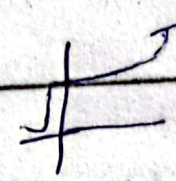
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→ Need:- if act. function not used then model will not capture non-linear data.

→ Neural network will behave like a linear model.

ideal act. function:-

1) Non linear $y = f(x)$ Linear x non linear ✓

e.g :- $\sigma(z) = \frac{1}{1 + e^{-z}}$ 

2) Differentiable $\rightarrow$ gradient des $\rightarrow$ derivatives.

3) Computationally inexpensive:-

$\rightarrow$ derivative $\rightarrow$ simple, easy, fast

if not training slow

4) Zero centered:- $\rightarrow$ output $\rightarrow$ zero centered

$\rightarrow$ normalize.

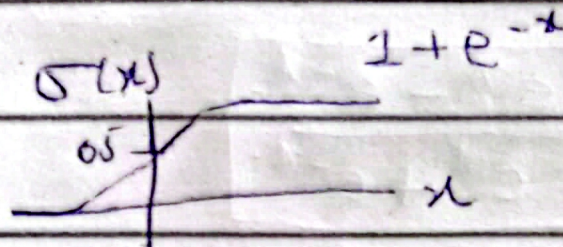
5) Non saturating:-

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Sigmoid function:-

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



Advantages:-

→ Output b/w 0 & 1 → Probability →

Output layer → Binary classification

→ Non linear → N.L data pattern capture.

→ Differentiable → Backprop → $\frac{\partial L}{\partial w}$

Disadvantages:-

1) Saturating function → Squeeze input

Van gradient problem.

2) Non zero centered → change is →

either +ve, -ve

so training b/m slow.

3) $\sigma(2) = \frac{1}{1 + e^{-2}}$

computation problem

b/c of exponent.

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Tanh Act. function :-

b/w -1 to 1

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$f(x) = 1 - \tanh^2(x)$$

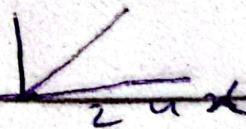
Advantages:-

- Non linear.
- Differentiable.
- Zero centered. $\left. \begin{matrix} \text{+ve} \\ \text{-ve} \end{matrix} \right\} \text{training faster}$

Dis:- → Saturating.

→ Computationally expensive.

Relu Act func:-



$$f(x) = \max(0, x)$$

Adv:-

Non linear,

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→ Not saturated in +ve reg.

→ Computational inexpensive. → faster converg

Dis:-

$$\text{Diff} \rightarrow \begin{matrix} x < 0 \rightarrow 0 \\ x \geq 0 \rightarrow 1 \end{matrix}$$

→ Not zero centered. → Solve by
Batch normalized

Dying relu problem:-

Sometimes for any input, output of a neuron is 0, called dead neuron, but its always 0.

→ If it increase from 50% to 100%, doesn't catch patterns

$$W_1 = W_1 - \eta \left(\frac{\partial L}{\partial W_1} \right) \leftarrow 0$$
$$W_1 = W_{old}$$

Reason behind that b/c 0 or negative:-

- 1) High learning rate. (η)
- 2) High -ve bias.

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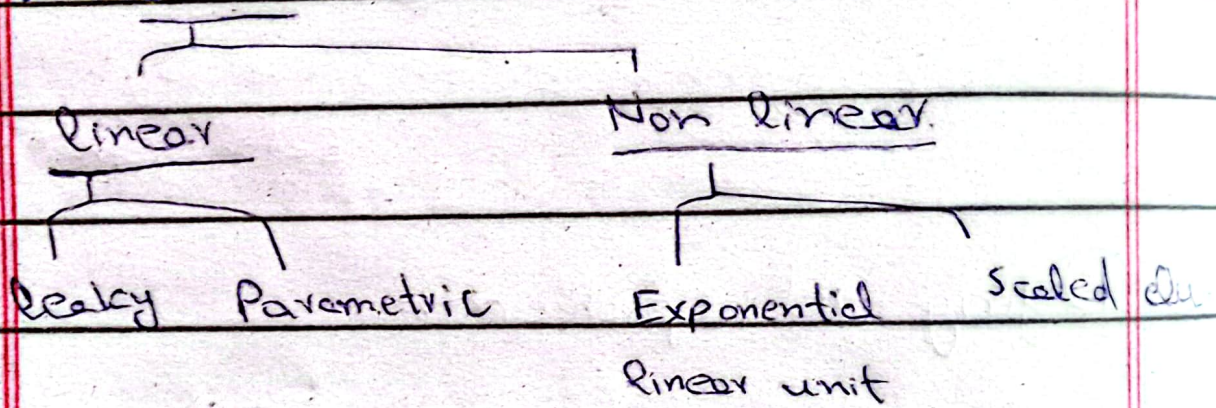
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Sol:- $\rightarrow$ Set low learning rate

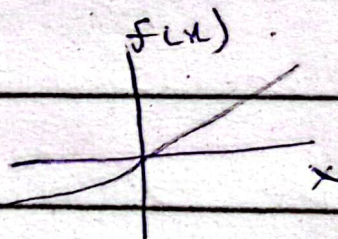
$\rightarrow$ Bias $\rightarrow$ +ve value $\rightarrow$ 0.01 $\rightarrow$ Experimented proven.

$\rightarrow$ Don't use relu $\rightarrow$ variants.

Relu variants:-



Leaky Relu:-



$$f(z) = \max(0.01z, z)$$

$$z \geq 0 \rightarrow z$$

$$z < 0 \rightarrow 1(z)$$

10%

output

$$\text{-ve } (z) \rightarrow \neq 0 \text{ } \boxed{0.01} \text{ of } z$$

So now derivative of update rule will not be zero.

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Non saturated. $\rightarrow$ Easily computed

No dying relu problem $\rightarrow$ close to 0 centered

Dis-advan:-

Why we take value as 0.01.

Parametric relu:-

$$f(x) = x \text{ if } x > 0.$$

ax otherwise.

$\swarrow$
trainable parameter.

depends on data a value find out

so we can say flexibility come.

Exponential linear unit:-

$$\text{Elu} = \begin{cases} x & \text{if } x > 0. \\ \alpha(e^x - 1) & \text{if } x < 0 \end{cases}$$

$\swarrow$
const $\rightarrow$ value chosen from

0.2 to 0.3.

Advantages:-

$\rightarrow$ Close to 0 centered $\rightarrow$ value.

$\hookrightarrow$ Convergence faster.

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- Better generalized results.
- Always continuous & differentiable.

Dis :-

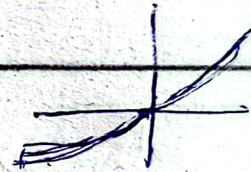
- Computational expensive $\rightarrow e^x$.

Selu :-

$$\text{Selu}(x) = \frac{1}{a} \begin{cases} x & \text{if } x > 0 \\ a e^{x/a} & \text{if } x < 0 \end{cases}$$

$$a \approx 1.673$$

$$\frac{1}{a} \approx 1.050$$



Benefits:-

Self normalizing $\rightarrow$ act
 $\downarrow$
normalized.

$$\boxed{m=0, \sigma=1.}$$

Converges faster.

Dis:-

Research not completed yet.